MSG No. 125.370 $P4'/n'bm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'2'2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[5,6,7,8,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'2'2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 3: Wyckoff site: `2c`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,2,5,6,11,12,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[3,4,7,8,9,10,13,14]`

Table 4: Wyckoff site: `2d`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,2,5,6,11,12,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[3,4,7,8,9,10,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `..2/m'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3,13,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,4,14,16]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[5,8,9,12]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[6,7,10,11]`

Table 6: Wyckoff site: `4f`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,3,13,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,4,14,16]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,8,9,12]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,7,10,11]`

Table 7: Wyckoff site: `4g`, site symmetry: `4'..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2,9,10]`
2	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[3,4,11,12]`
3	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,6,13,14]`
4	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[7,8,15,16]`

Table 8: Wyckoff site: `4h`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,2,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[3,4,13,14]`
3	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[7,8,9,10]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2`

No.	position	mapping
1	$[x, x, 0]$	`[1,3]`
2	$\left[\frac{1}{2} - x, \frac{1}{2} - x, 0\right]$	`[2,4]`
3	$\left[x + \frac{1}{2}, -x, 0\right]$	`[5,8]`
4	$\left[-x, x + \frac{1}{2}, 0\right]$	`[6,7]`
5	$\left[\frac{1}{2} - x, x, 0\right]$	`[9,12]`
6	$\left[x, \frac{1}{2} - x, 0\right]$	`[10,11]`
7	$[-x, -x, 0]$	`[13,15]`
8	$\left[x + \frac{1}{2}, x + \frac{1}{2}, 0\right]$	`[14,16]`

Table 10: Wyckoff site: `8j`, site symmetry: `..2`

No.	position	mapping
1	$\left[x, x, \frac{1}{2}\right]$	`[1,3]`
2	$\left[\frac{1}{2} - x, \frac{1}{2} - x, \frac{1}{2}\right]$	`[2,4]`
3	$\left[x + \frac{1}{2}, -x, \frac{1}{2}\right]$	`[5,8]`
4	$\left[-x, x + \frac{1}{2}, \frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{2} - x, x, \frac{1}{2}\right]$	`[9,12]`
6	$\left[x, \frac{1}{2} - x, \frac{1}{2}\right]$	`[10,11]`
7	$\left[-x, -x, \frac{1}{2}\right]$	`[13,15]`
8	$\left[x + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2}\right]$	`[14,16]`

Table 11: Wyckoff site: `8k`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x, \frac{1}{4}, 0\right]$	`[1,11]`
2	$\left[\frac{1}{2} - x, \frac{1}{4}, 0\right]$	`[2,12]`

continued ...

Table 11

No.	position	mapping
3	$\left[\frac{1}{4},\, x,\, 0\right]$	[3,9]
4	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, 0\right]$	[4,10]
5	$\left[\frac{3}{4},\, -x,\, 0\right]$	[5,15]
6	$\left[\frac{3}{4},\, x+\frac{1}{2},\, 0\right]$	[6,16]
7	$\left[-x,\, \frac{3}{4},\, 0\right]$	[7,13]
8	$\left[x+\frac{1}{2},\, \frac{3}{4},\, 0\right]$	[8,14]

Table 12: Wyckoff site: 8l, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, \frac{1}{2}\right]$	[1,11]
2	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{2}\right]$	[2,12]
3	$\left[\frac{1}{4},\, x,\, \frac{1}{2}\right]$	[3,9]
4	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[4,10]
5	$\left[\frac{3}{4},\, -x,\, \frac{1}{2}\right]$	[5,15]
6	$\left[\frac{3}{4},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[6,16]
7	$\left[-x,\, \frac{3}{4},\, \frac{1}{2}\right]$	[7,13]
8	$\left[x+\frac{1}{2},\, \frac{3}{4},\, \frac{1}{2}\right]$	[8,14]

Table 13: Wyckoff site: 8m, site symmetry: ..m'

No.	position	mapping
1	$[x,\, -x,\, z]$	[1,15]
2	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, z\right]$	[2,16]
3	$[-x,\, x,\, -z]$	[3,13]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$	[4,14]
5	$\left[\frac{1}{2}-x,\, -x,\, -z\right]$	[5,12]
6	$\left[x,\, x+\frac{1}{2},\, -z\right]$	[6,11]
7	$\left[-x,\, \frac{1}{2}-x,\, z\right]$	[7,10]
8	$\left[x+\frac{1}{2},\, x,\, z\right]$	[8,9]

Table 14: Wyckoff site: 16n, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[2]
3	$[y,\, x,\, -z]$	[3]
4	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[4]
5	$\left[y+\frac{1}{2},\, -x,\, -z\right]$	[5]
6	$\left[-y,\, x+\frac{1}{2},\, -z\right]$	[6]
7	$\left[-x,\, y+\frac{1}{2},\, z\right]$	[7]
8	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[8]
9	$\left[\frac{1}{2}-y,\, x,\, z\right]$	[9]

continued ...

Table 14

No.	position	mapping
10	$\left[y,\, \frac{1}{2} - x,\, z\right]$	[10]
11	$\left[x,\, \frac{1}{2} - y,\, -z\right]$	[11]
12	$\left[\frac{1}{2} - x,\, y,\, -z\right]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$\left[x + \frac{1}{2},\, y + \frac{1}{2},\, -z\right]$	[14]
15	$[-y,\, -x,\, z]$	[15]
16	$\left[y + \frac{1}{2},\, x + \frac{1}{2},\, z\right]$	[16]